CS23532 - COMPUTER NETWORKS LABORATORY

PRACTICAL EXERCISE 2: Physical Layer Media & Custom Twisted-Pair Cable Assembly

1. OBJECTIVE AND AIM

To acquire a comprehensive understanding of physical layer guided transmission media (network cables) and to independently fabricate, crimp, and test customized Ethernet Straight-Through and Crossover patch cables using RJ-45 connectors according to T568A and T568B wiring standards.

2. SECTION A: OVERVIEW OF GUIDED TRANSMISSION MEDIA

Physical layer communication relies on guided media to convey electrical or optical signals across nodes. The four primary classes of network cabling are detailed below:

Cable Type	Specification / Category	Max Data Rate & Bandwidth	Key Advantages & Limitations	Primary Network Applications
Unshielded Twisted Pair (UTP)	Cat 3 / Cat 5e / Cat 6	10 Mbps to 1 Gbps (100 MHz - 250 MHz)	**Advantages:** Cost-effective, high flexibility, easy installation. **Disadvantages:** Highly vulnerable to Electromagnetic Interference (EMI) and crosstalk over distance.	Standard Office LANs, Legacy Telephony, Workstation-to-Switch drops.
Shielded Twisted Pair (STP / S/FTP)	Cat 6a / Cat 7 / Cat 8	10 Gbps to 40 Gbps (500 MHz - 2000 MHz)	**Advantages:** Foil/braided shielding provides superior protection	Data centers, industrial automated environments,

Cable Type	Specification / Category	Max Data Rate & Bandwidth	Key Advantages & Limitations	Primary Network Applications
			against EMI/RFI and alien crosstalk. **Disadvantages:** Higher cost, rigid bend radius, requires proper grounding.	high-density server racks.
Coaxial Cable	RG-6 / RG-59 / RG-11	10 Mbps to 1 Gbps (Broadband)	**Advantages:** Solid copper core provides strong signal isolation over longer distances than UTP. **Disadvantages:** Bulkier, difficult to pull through conduit, single point of failure in bus topologies.	Cable television networks (CATV), Broadband ISP connections, CCTV camera feeds.
Fiber Optic Cable	Single-Mode (SMF) & Multi-Mode (MMF)	10 Gbps to 100+ Gbps (Terahertz light signals)	**Advantages:** Complete immunity to EMI, ultra-low attenuation, extremely long range (up to 40+ km). **Disadvantages:** High initial investment, fragile glass core, requires specialized optical splicing tools.	Inter-building backbones, Subsea telecommunications, High-speed Metro Ethernet.

3. SECTION B: FABRICATION OF

STRAIGHT-THROUGH AND CROSSOVER CABLES

3.1 Required Tooling and Materials

- **Four-pair Twisted-Pair Ethernet Cable:** Category 5e or Category 6 standard.
- **RJ-45 Modular Connectors:** Standard 8P8C (8 Position, 8 Contact) plugs.
- **Ratchet Crimping Tool:** Integrated blade assemblies for jacket stripping, wire trimming, and pin compression.
- **Protective Strain-Relief Boots:** Rubber boots to protect the locking clip.
- **Continuity Network Cable Tester:** To verify wire pinouts across both ends.

3.2 Pinout Configuration Standards (TIA/EIA-568)

The standard color-coding for Ethernet connections is defined under T568A and T568B specifications:

Pin Number	Signal Function	T568A Wire Color	T568B Wire Color
Pin 1	Transmit (+) / Receive (+)	White / Green	White / Orange
Pin 2	Transmit (-) / Receive (-)	Green	Orange
Pin 3	Receive (+) / Transmit (+)	White / Orange	White / Green
Pin 4	Unused / Power	Blue	Blue
Pin 5	Unused / Power	White / Blue	White / Blue
Pin 6	Receive (-) / Transmit (-)	Orange	Green
Pin 7	Unused / Power	White / Brown	White / Brown

Pin Number	Signal Function	T568A Wire Color	T568B Wire Color
Pin 8	Unused / Power	Brown	Brown

Cable Type Definitions:
• **Straight-Through Cable:** Configured identically on both ends (End A: T568B & End B: T568B). Used for connecting MDI to MDI-X devices (e.g., PC to Switch).

• **Crossover Cable:** Configured with T568A on one end and T568B on the other. Used for connecting directly between similar MDI devices (e.g., PC to PC).

3.3 Step-by-Step Cable Assembly Procedure

1. **Protective Boot Insertion:** Slide the strain-relief rubber boots onto each end of the cable facing outwards prior to stripping.
2. **Outer Jacket Stripping:** Utilize the stripper blade on the crimping tool to remove approximately 1.5 cm (0.6 inches) of the outer PVC jacket without nicking the inner copper wire insulation.
3. **Unraveling and Alignment:** Untwist the four wire pairs. Straighten and align the individual conductors side-by-side in exact compliance with the selected TIA/EIA standard (T568A or T568B).
4. **Trimming Conductor Length:** Firmly squeeze the aligned wires together and trim them cleanly perpendicular using the crimper's flat cutting blade, leaving ~1.2 cm of exposed conductor.
5. **RJ-45 Plug Insertion:** Hold the RJ-45 connector with its retaining clip facing downward and gold pins facing upward. Carefully push the conductors straight into the plug channels until all wires hit the front tip and the outer sheath sits inside the plug's retention clamp zone.
6. **Mechanical Crimping:** Insert the loaded RJ-45 plug into the 8P8C crimping cavity. Squeeze the handles firmly until the ratchet mechanism releases, driving the gold-plated teeth into the conductors and locking the cable sheath.
7. **Secondary Termination & Verification:** Repeat Steps 2 through 6 on the opposite cable end (matching standard for Straight-Through, alternating standard for Crossover). Test the completed assembly using a modular LAN cable tester.

4. SECTION C: LABORATORY OBSERVATIONS AND

ANALYTICAL WORKBOOK

Observation Question 1: What is the functional architectural difference between a Crossover cable and a Straight-Through cable?

Solution / Answer: A Straight-Through cable maintains identical pin mappings on both terminations (T568B to T568B), ensuring signal paths (TX to TX, RX to RX) are mapped to be crossed inside intermediate MDI-X switch ports. A Crossover cable internally transposes the Transmit pins (1 & 2) on one end to the Receive pins (3 & 6) on the opposite end (T568A to T568B), allowing two direct interfaces (MDI to MDI) to communicate without an intervening hub or switch.

Observation Question 2: Which cable type is required to connect two host computers directly via Ethernet network interface cards without Auto-MDIX enabled?

Solution / Answer: A **Crossover Cable** is required because both PC Network Interface Cards (NICs) operate as MDI devices transmitting on Pins 1 & 2 and listening on Pins 3 & 6. The crossover wiring ensures Transmit output connects directly to the opposite PC's Receive input.

Observation Question 3: Which cable configuration is used to connect a Personal Computer (PC) to an unmanaged Ethernet switch port?

Solution / Answer: A **Straight-Through Cable** is used. The host PC (MDI) connects to the switch port (MDI-X), which internally swaps the receive and transmit circuits.

Observation Question 4: Identify the specific category of twisted pair cabling installed in your laboratory infrastructure and its rated capability.

Solution / Answer: The laboratory is equipped with **Category 6 (Cat 6) UTP cabling**, which supports data transmission speeds up to 10 Gbps (for distances up to 55 meters) with a bandwidth frequency of 250 MHz.

Observation Question 5: Document your experimental findings, operational challenges encountered during fabrication, and final testing results.

Solution / Answer:
• **Results:** Both Straight-Through and Crossover cable assemblies successfully passed the LED sequential continuity test on the LAN cable tester (LEDs 1 through 8 illuminated in

order for Straight-Through; 1-3, 2-6, 3-1, 6-2 crossed pattern for Crossover). Direct PC-to-PC link was verified at 1 Gbps.

• **Practical Challenges & Solutions:**

1. *Wire Disalignment during Insertion:* Individual conductors untwisted and crossed paths inside the RJ-45 body. Solved by flattening and cropping conductors cleanly in a tight row before pushing firmly.
2. *Insufficient Jacket Retention:* Stripping too much outer jacket caused the crimp clamp to miss the sheath. Corrected by shortening exposed wire lengths to ~1.2 cm so the outer jacket locks inside the connector body for proper strain relief.